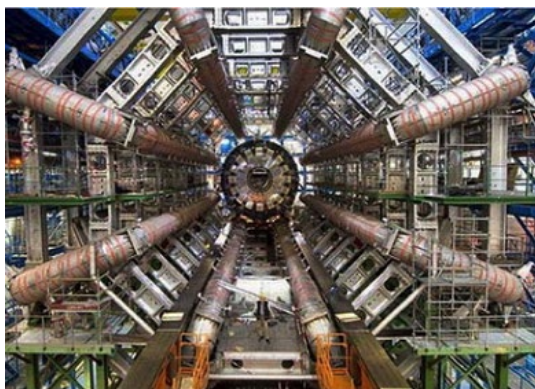
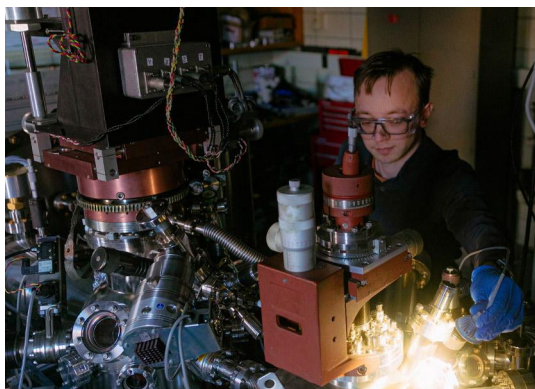
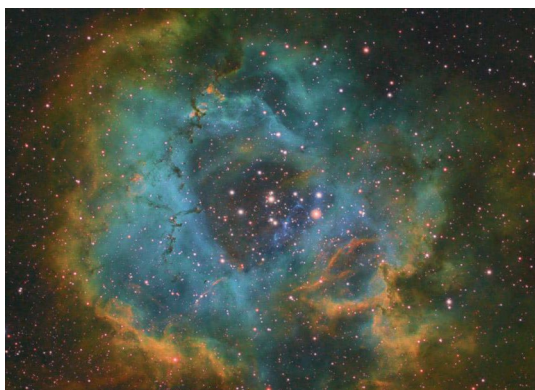


Department of Physics and Astronomy

Iowa State University



Graduate Handbook

Revised Fall 2025

Introduction

To complete the MS or PhD program at ISU you must successfully satisfy two sets of requirements, namely, the Graduate College requirements and the requirements of the Department of Physics and Astronomy. The first set of requirements appears in the *Graduate College Handbook*, which can be found online at grad-college.iastate.edu/handbook. The second set of requirements is presented in this document, which we hope will aid you and your advisor in making the decisions that will affect your graduate studies. Please look this document over carefully.

If you have questions about the program, please contact the first-year advisor or your major professor. Questions can also be directed to the Graduate Secretary, the Director of Graduate Education (DOGE) or any member of the Graduate Status Committee. If you are uncertain about any aspect of our graduate program, please ask – it is better to ask now than to discover problems later.

Department Tuition Scholarships and Teaching Assistant Stipends

The department offers scholarships covering the portion of tuition not covered by the College of Liberal Arts and Sciences to all incoming students. The continuation of this scholarship beyond the first year is contingent upon both satisfactory academic and teaching performance.

Graduate students who are making satisfactory academic progress are usually appointed as teaching assistants for up to two academic years. After the first semester, the stipend for this appointment will be based on teaching performance.

Oral English Certification Test

Graduate students for whom English is not their native language are required to take the University-administered Oral English Certification Test (OECT) to qualify for teaching assignments within the department. Your certification level on this exam determines what type of teaching assignments you may be given. Students who do not qualify at Level 1 on the OECT will be required to take GRST 5400 to improve your English skills.

The department requires that you make progress towards at least Level-2 certification during your first year. You are required to achieve at least Level-3 certification by the end of your first semester at ISU, and at least Level-2 certification by the beginning of the fall semester of your second year. Your tuition scholarship is tied in part to your teaching performance and your Oral English Certification is a part of that performance. Failure to achieve the required level of certification may result in the loss of the tuition scholarship.

Advanced students may also return to teaching after joining a research group. In this case you must have achieved at least a Level-2 certification or you will not receive a department contribution towards the tuition scholarship.

Graduate Degrees and Majors

The department offers a PhD degree along with thesis and non-thesis MS degrees. Five graduate majors are available: Astrophysics (AST), Condensed Matter Physics (CMP), High Energy Physics (HEP), Nuclear Physics (NUC), and Physics (PHY). The choice of graduate major affects course requirements and Academic Plan Committee membership, and students should choose a major in consultation with their major professor. Departmental degree requirements are summarized below, and the Graduate Student Handbook should be consulted for additional requirements regarding items such as transfer credits, time limits, and residency requirements.

MS Degree (thesis option)

The required courses for PHY, CMP, HEP and NUC majors are Physics 5710, 5720, 5910, 5920 and either 5310 or 5640. The required courses for AST majors are Astronomy 5050, Astronomy 5800 or 5840, and two of Physics 5170, 5910, 5310 and 5640. Physics 5510 is strongly recommended for all MS students. Since Astronomy 5800 and 5840 are offered irregularly, a different graduate-level course may be substituted with DOGE approval if necessary.

A minimum of 30 graduate-level credits are required in total. At least 21 credits must be from courses in the Department of Physics and Astronomy, and at least 15 of these credits must be in courses at the 5000 and 6000 level. Up to 8 research credits of 6990 (but not 5990) can be applied toward the 30-credit minimum. Students are expected to research and write a thesis that demonstrates independent and creative work.

MS Degree (non-thesis option)

Requirements are the same as the thesis option except that the research component is met through a maximum of two credits of 5990. No credits of 6990 can be counted toward the total of 30 credits. A written document (e.g., a literature review, an annotated bibliography, or a project report) demonstrating substantial individual accomplishment outside of coursework must be prepared.

MS Degree – General Considerations

MS students must present their thesis or creative component as part of a final oral examination.

MS students should work with their major professor on choosing courses that best fit their future career and education plans.

The department does not admit students directly into the MS program. If a student needs to move to the MS from the PhD, the student and their major professor should inform the Graduate Secretary as soon as possible for guidance regarding funding and Graduate College deadlines.

PhD Degree – Course Requirements

A minimum of 72 graduate credits must be earned for the PhD. At least 36 credits, including all the dissertation research credits, must be earned under the supervision of the student's Academic Plan Committee.

Required courses for PHY, CMP, HEP, and NUC majors are Physics 5710, 5720, 5910, 5920, 5310 and 5640. The required courses for AST majors are Astronomy 5050, 5800 and 5840, Physics 5170, 5910, and either Physics 5310 or 5640. Since Astronomy 5800 and 5840 are offered irregularly, a different graduate-level course may be substituted with DOGE approval if necessary. Physics 5510 is strongly recommended for all PhD students.

To avoid overspecialization, students in all graduate majors must take a significant body of pertinent coursework outside the major field. For PHY majors, the judgment of what constitutes outside the major field is decided by the student's major professor in consultation with the DOGE. At a minimum, 9 credits of applicable graduate courses are required (either offered inside or outside the department) and they must include at least one 5000- or 6000-level introductory course in another area of physics (see Appendix A). Courses in other departments must be listed as "non-major graduate credit" in the ISU course catalog.

PhD Degree – Course Waivers & Transfer Credits

Students with prior graduate-level coursework can make a request to the Graduate Status Committee (GSC) for a required course waiver. Students should first meet with the GSC chair to discuss their situation. A waiver request must demonstrate that the course they have taken elsewhere is equivalent, in both topics and level, to the ISU course in question, and that a grade equivalent to an A- or A at ISU was awarded.

In rare cases students may wish to transfer credits from another institution to ISU. This is a different process than a course waiver and requires approval by both the department and the Graduate College. As with a waiver, the first step should be a meeting with the GSC chair.

PhD Degree – Teaching Requirements

All PhD students are required to teach one year of elementary physics and/or astronomy. Only teaching in at least one recitation section per semester is counted toward this requirement, labs and grading assignments do not. Please note that your level of Speak-Teach (OECT) certification will determine what sort of assignments you can be offered.

PhD Degree – Qualifying Exams

All PhD students are required to pass two qualifying exams. One exam, given annually in August, focuses on classical physics, and the other exam, given annually in January, focuses on modern physics. A document describing the exam procedures and regulations in detail can be obtained from the Graduate Secretary.

Choosing your Major Professor (Advisor)

Your relationship with your major professor is critical to your success as a graduate student, and you are encouraged to talk with potential advisors early in your first year in the department. Ideally you should have a major professor in place by the start of your second year.

To learn about the various research groups be sure to take advantage of the Introductory Seminar presentations and talk to more senior graduate students about their work and their experiences. Each research group is run differently, and your goal is to find both an area of research that excites you and a research group that you can see yourself being part of.

The Graduate College has prepared a Student & Major Professor Checklist available at: www.grad-college.iastate.edu/student#you&yourprofessor. You are encouraged to use this resource as a starting point for a discussion with potential major professors.

Academic Plan and Academic Plan Committee

Both MS and PhD students must submit an Academic Plan (AP) and form an Academic Plan Committee (APC) once they have selected a major professor. For PhD students, both the AP and the APC must be in place before the student can take their preliminary oral exam. For MS students they must be in place before the final exam. Your major professor and the Graduate Secretary can assist you with preparing the AP and APC in Workday, and extensive information about both the AP and APC can be found in the Graduate College Handbook.

The AP is a listing of all courses that you have taken (or plan to take) to fulfill your degree requirements. You may take additional courses at ISU, but only those taken to fulfill your degree should be listed on the AP. The department requires that the GPA for the courses listed on the AP be 3.0 or greater. The AP will be examined by the DOGE and by the Graduate College to make sure that it meets both departmental and university requirements.

The APC is the committee that will oversee your thesis research. For PhD students the APC consists of five graduate faculty members, while for MS students the APC has three members. The APC must include your major professor and one faculty member outside of your area of research. A typical PhD APC consists of your major professor, two faculty members from the department in your area of research, one faculty member from the department outside your area of research, and another faculty member from a different department. A typical MS APC consists of your major professor, one faculty member in your area of research, and one faculty member outside of your area of research.

PhD Degree – Preliminary Oral Exam

A PhD student who has passed both written Qualifying Exams must pass a Preliminary Oral Exam before the final Thesis defense. The preliminary oral exam should be taken within one year of the appointment of the student's Academic Plan Committee. If applicable, students must reach at least Level 2 OECT certification before taking the Preliminary Oral Exam.

The purpose of the Preliminary Oral Exam is to demonstrate to the student's advisor and committee that the student is: 1) ready to undertake a PhD research project and has a mastery of general physics/astronomy; and 2) to assist the student and advisor in planning and completing the thesis research project in a timely fashion. A 30-50 minute talk should be prepared by the student on a topic decided mutually with the advisor. It is beneficial for the selected topic to be related to the student's future PhD thesis research field; however, original research results are not required for the Preliminary Oral Exam. For example, the talk may review a key paper from the literature; it may describe the student's expertise in using relevant laboratory equipment, or it may demonstrate the student's analytical or computational competence required for their research. The talk may also describe, for example, a future research problem that could be the basis of the PhD thesis, with the understanding that the final thesis may deviate from the original plans, based on information gained as the project progresses.

The evaluation of the Preliminary Oral Exam by the committee depends on the understanding demonstrated to the committee by the student of the methodology of carrying out PhD work for a specific problem and not on the amount of research work completed at the time of the Preliminary Oral Exam. Questions on general physics and astronomy knowledge are a normal part of the Preliminary Oral Exam. For these reasons, this exam should be taken without a significant lapse of time after passing the Qualifying Exams.

Graduate Status Committee Annual Student Review

For first-year students the GSC reviews student grades at the end of the first semester to ensure that students are making satisfactory progress in their courses (GPA 3.0 or higher). Students with a low GPA will meet with the first-year advisor to develop an appropriate plan for the spring semester aimed at improving academic performance.

The academic and research performance of all students is evaluated by the GSC at the end of each academic year. Feedback from both students and advisors is solicited by the Graduate Secretary early in the spring semester as part of this process. The main concern of the GSC is to ensure that students are maintaining an acceptable level of academic performance (3.0 GPA or better), and that they are on track to graduate within six years (for the PhD) or three years (for the MS).

Guidelines for the minimum level of academic performance are:

- A student is expected to earn at least six credits per semester with a GPA of at least 3.0 during the first year as a graduate student.
- After two years, a student is expected to have earned a total of at least 24

credits (17 of which should be in 5000- and 6000-level physics or astronomy courses) with a GPA of at least 3.0.

- Students beyond the second year of graduate study must maintain a cumulative GPA at or above 3.0.

Note that research and seminar credits are not used in computing the GPA, and students with transfer credits are handled on an individual basis. A student whose performance falls below these standards risks loss of financial support and may be declared ineligible for further work toward an advanced degree in the department. Written comments from the student's faculty advisor will be considered before a decision is reached on the student's status. A student may petition the GSC for special consideration in unusual circumstances.

Guidelines for a reasonable PhD timeline are:

- Both qualifying exams should be passed by the end of Year 2.
- A major professor should be named by the end of Year 2.
- The AP and APC should be in place by the end of the fall semester of Year 3.
- The Oral Preliminary exam should be passed within one year of APC formation.
- Research should be planned with the goal of graduation at the end of Year 6.

The exact timeline for MS students will vary depending on when they have switched from the PhD to the MS degree. The department expects all MS students to complete the degree requirements in three years, including time spent as a PhD student.

Graduation

There are a number of important Graduate College deadlines associated with graduation. As you get closer to your graduation date, please pay careful attention to any communications you receive from either the Graduate College or the departmental Graduate Secretary.

Students are strongly encouraged to talk with the Graduate Secretary at the start of the academic year in which they plan to graduate. Both students and supervisors are encouraged to review the Graduate College Handbook chapter on Degree Completion.

Appendix A: Outside Area Course Listings by Graduate Major

Courses - Physics	Astrophysics	High Energy Physics	Nuclear Physics	Condensed Matter Physics
5010 (Oral Comm.)				
5020 (Intro. Sem.)				
5110/20 (CMP I/II)	OUTSIDE	OUTSIDE	OUTSIDE	
5220 (Quant. Comp.)	OUTSIDE	OUTSIDE	OUTSIDE	OUTSIDE
5260 (Part. and Nuc.)	OUTSIDE			OUTSIDE
5280 (Math Methods)	OUTSIDE	OUTSIDE	OUTSIDE	OUTSIDE
5310 (Stat. Mech.)				
5340 (Symmetry. and Group Theory)	OUTSIDE	OUTSIDE	OUTSIDE	OUTSIDE
5350/60 (SC Devices)	OUTSIDE	OUTSIDE	OUTSIDE	
5410 (Gen. Relativity)	OUTSIDE	OUTSIDE	OUTSIDE	OUTSIDE
5510 (Comp. Physics)				
5640 (Mechanics)				
5710/20 (E&M)				
5910/20 (QM)				
6110 (Quantum CM)	OUTSIDE	OUTSIDE	OUTSIDE	
6240/50 (Adv. Nuc.)	OUTSIDE	OUTSIDE		OUTSIDE
6370/80 (Adv. HEP)	OUTSIDE		OUTSIDE	OUTSIDE
6810 (QFT I)	OUTSIDE			
6820 (QFT II)	OUTSIDE	OUTSIDE	OUTSIDE	OUTSIDE
Courses - Astronomy				
5050 (Cosmology)		OUTSIDE	OUTSIDE	OUTSIDE
5800 (Stellar)		OUTSIDE	OUTSIDE	OUTSIDE
5840 (Gal. and Extragal.)		OUTSIDE	OUTSIDE	OUTSIDE

Updated: Spring 2025

Notes

1. A blank means a course cannot be used as an outside-of-area credit for that graduate major.
2. For PHY majors, the judgment of what courses are outside-of-area is decided by the student's major professor in consultation with the DOGE.
3. Special Topics (5900 and 6600) and Advanced Seminar (6500) courses in a particular major can be counted as outside-of-area credits for the other majors.

Appendix B: Graduate Minor Requirements

A graduate student seeking a **PhD with a minor in physics** is required to have:

- at least 12 credits in physics or astronomy courses acceptable for graduate credit.
- of these 12 at least 6 credits must be from the following list of (core) courses: Phys 4800, 4810, 5710, 5720, 5910, 5920, 5310, 5640.

A graduate student seeking an **MS with a minor in physics** is required to have:

- at least 6 credits in physics or astronomy courses acceptable for graduate credit.
- of these 6 at least 3 credits must be from the following list of (core) courses: Phys 4800, 4810, 5710, 5720, 5910, 5920, 5310, 5640

Note, the department does not offer specific graduate minors corresponding to the graduate majors in astrophysics, condensed matter physics, nuclear physics and high energy physics.

Students are reminded that the Graduate College requires them to have an APC member from the minor area. Please see the Graduate Student Handbook for more details.

Appendix C: Graduate Co-Major Requirements

A graduate student seeking co-major MS or PhD degree in Physics or Astronomy must fulfill the same requirements as for the respective regular degree with exception of “out of area” course requirements. Specifically, they must:

- have a co-advising faculty from Department of Physics and Astronomy
- complete the required core courses
- pass both the qualifying and preliminary exams (PhD only)
- pass a thesis defense examination. The thesis defense is a joint exam between the two Departments. Selection of the APC should provide adequate representation of the faculty members from both departments, including an "out-of-area" member for each department.

The "out of area" course requirement is naturally fulfilled by the courses taken as the other part of the co-major degree.”

Approved by Department Faculty in October 2015.

Minor edits and updates by Department DOGE in April 2025.